EE 671: VLSI DESIGN

PROJECT REPORT -1

Team Information

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Problem Statement

Circuit Name	Description	Function
dfxtn	Delay flop, single output Q, negedge clk	---
o21bai	2-input OR into first input of 2-input NAND, 2nd input	Y = !((A1 | A2) & !B1_N)

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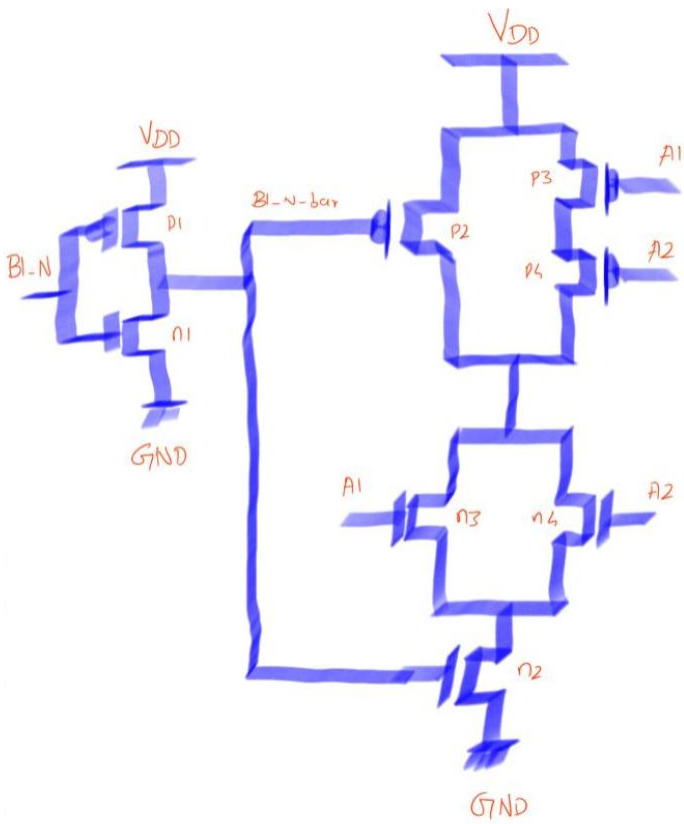
1. Circuit Diagram of the Cell with Dimension

Circuit Diagram of the Cell with a table of MOSFET width and length.

1.1 Combinational Circuit

Y = !((A1 | A2) & !B1_N)



Size Table:

L=0.42u

Width of Transistors are as follows -

P1	P2	P3	P4	N1	N2	N3	N4
5.08u	3.81u	8.89u	2.54u	2.1u	4.2u	1.26u	1.26u

Note: For the Combinational Circuit, the worst case tr and tf for the above W and L sizes were found to be –

tr = 28 ps (approx)

tf = 28 ps (approx)

However, one interesting behaviour was found -

If we tried to match the rise and fall time to that of Inverter (ie, tr = 19.3 ps and tf = 19.3 ps (approx.)), the following sizes were calculated after repetitive experiments –

Size Table:

L=0.42u

Width of Transistors are as follows -

P1	P2	P3	P4	N1	N2	N3	N4
20.32u	21.59u	48.26u	12.7u	23.52u	24.78u	6.62u	6.62u

For the Combinational Circuit, the worst case tr and tf for the above W and L sizes were found to be –

tr = 22 ps (approx)

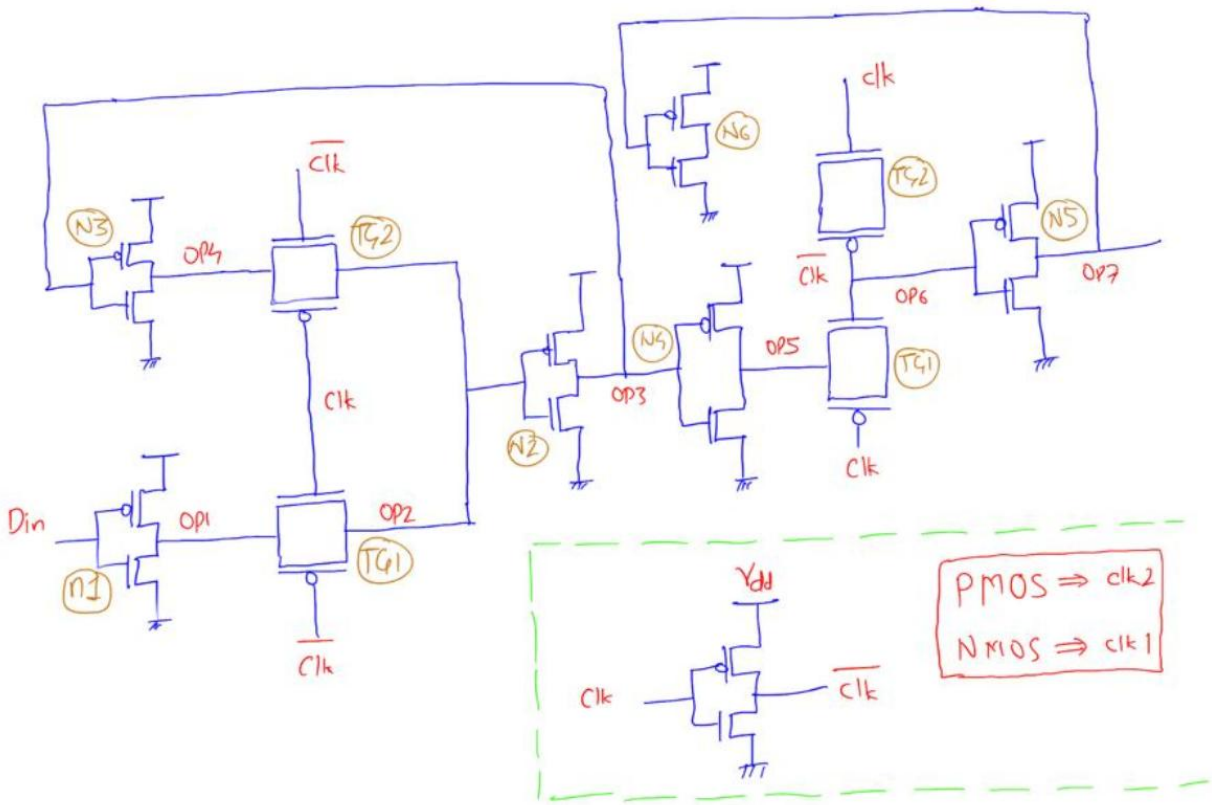
tf = 22 ps (approx)

A pattern was observed.

The decrease in rise/fall time was accompanied with the increase in transistor areas !!

1.2 Sequential Circuit

Delay flop, single output Q, negedge clk



Size Table:

L=0.42u

Width of the Transistors is as follows -

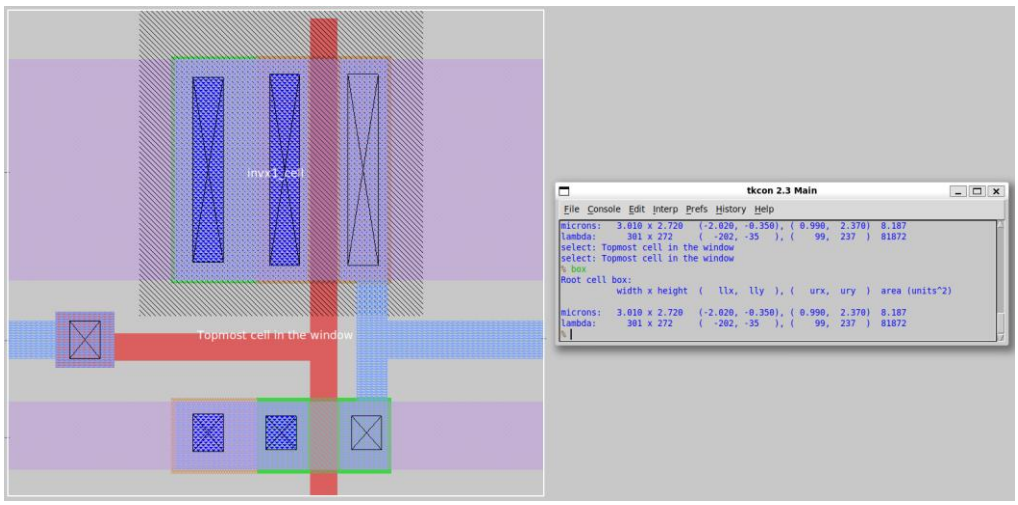
N1	N2	N3	N4	N5	N6	N7
WP: 1.9 WN: 1.5	WP: 1.25 WN: 1.2	WP: 0.5 WN: 0.5	WP: 2.48 WN: 1.77	WP: 2.75 WN: 1.5	WP: 0.5 WN: 0.42	WP: 0.5 WN: 0.42

TG1	TG2	N(CLK)
WP: 1.4 WN: 1.5	WP: 1.74 WN: 1.5	WP: 2 WN: 1.5

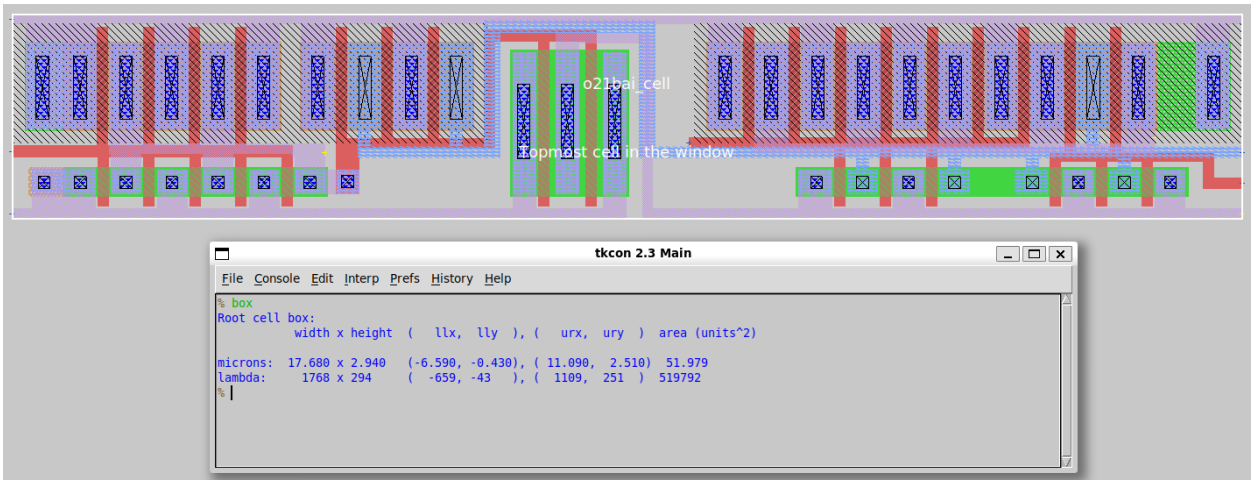
2. Screenshot of Cell Layout

Screenshot of the cell layout (with area mentioned as WIDTH X HEIGHT).

2.1 Combinational Circuit

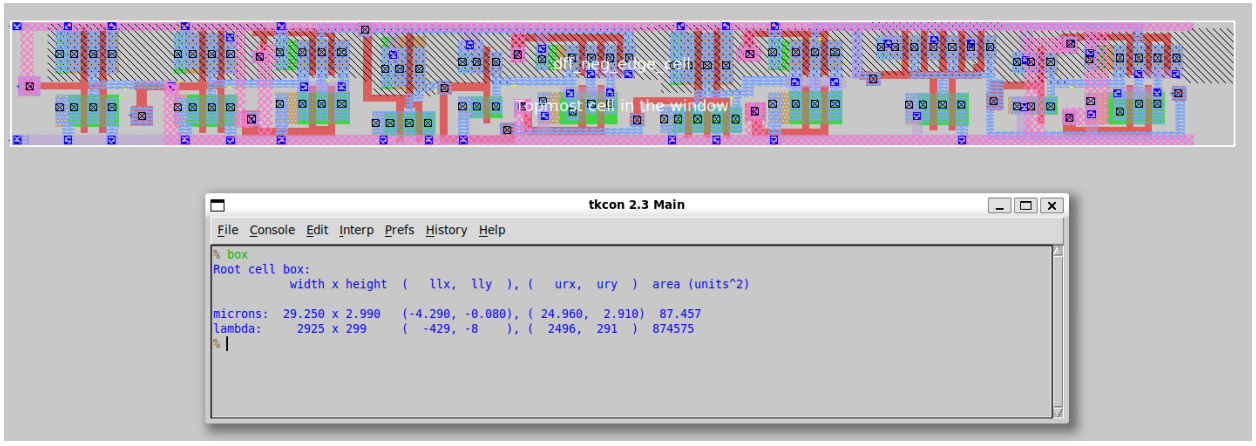


INVX1 Circuit Layout/Area



o21bai Circuit Layout/Area

2.2 Sequential Circuit



DFF Layout/Area

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3. Screenshot of the PEX Netlist

Screenshot of PEX Netlist.

3.1 Combinational Circuit

```
karunesh@Karunesh: ~/vlsi_d    karunesh@Karunesh: ~/vlsi_de
* NGSPICE file created from invx1_cell.ext - technology: sky130A

.subckt invx1_cell vdd gnd in out
X0 out in vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=3810,314 d=3810,314
X1 out in gnd gnd sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=1260,144 d=1260,144
C0 out in 0.03413f
C1 out vdd 0.22367f
C2 vdd in 0.13619f
C3 out gnd 0.27906f
C4 in gnd 0.40941f
C5 vdd gnd 0.84027f
.ends
```

INVX1 PEX

```
karunesh@Karunesh: ~/vlsi_d    karunesh@Karunesh: ~/vlsi_de
**devattr s=6350,354 d=3238,178
X26 p_common a2 out1 vdd sky130_fd_pr__pfet_01v8 ad=0.35983 pd=1.97778 as=0.38862 ps=2.136 w=1.27 l=0.15
**devattr s=3238,178 d=6477,356
X27 b1_n_bar b1_n gnd gnd sky130_fd_pr__nfet_01v8 ad=0.12852 pd=1.116 as=0.16604 ps=1.04933 w=0.42 l=0.15
**devattr s=1071,93 d=1071,93
X28 vdd a1 p_common vdd sky130_fd_pr__pfet_01v8 ad=0.41729 pd=2.29 as=0.35983 ps=1.97778 w=1.27 l=0.15
**devattr s=3238,178 d=3238,178
C0 vdd b1_n_bar 1.55443f
C1 p_common out1 0.62747f
C2 a2 n_common 0.05339f
C3 out1 b1_n 0.00275f
C4 a1 n_common 0.04383f
C5 out1 n_common 0.54695f
C6 p_common n_common 0.31706f
C7 a1 b1_n_bar 0.004f
C8 out1 b1_n_bar 0.22154f
C9 p_common b1_n_bar 0
C10 a2 vdd 0.16923f
C11 a1 vdd 0.58873f
C12 out1 vdd 0.84155f
C13 p_common vdd 2.1292f
C14 b1_n b1_n_bar 0.20219f
C15 b1_n_bar n_common 0.07512f
C16 vdd b1_n 0.41475f
C17 vdd n_common 0.30264f
C18 a2 a1 0.01863f
C19 a2 out1 0.13399f
C20 p_common a2 0.07021f
C21 a1 out1 0.16786f
C22 p_common a1 0.21434f
C23 a2 gnd 0.48756f
C24 a1 gnd 0.69343f
C25 out1 gnd 1.11245f
C26 b1_n gnd 0.74437f
C27 vdd gnd 4.25593f
C28 p_common gnd 0.11459f
C29 n_common gnd 1.52822f
C30 b1_n_bar gnd 1.58463f
.ends
```

o21bai PEX

3.2 Sequential Circuit

karunesh@Karunesh: ~/vlsi_d karunesh@Karunesh: ~/vlsi_de

```
C27 out_3 vdd 0.45167f
C28 Din clk 0.04241f
C29 clk out_4 0.11467f
C30 clk_bar out_5 0.17178f
C31 clk_bar out_7 0.08842f
C32 out_6 out_5 0.43979f
C33 out_6 out_7 0.32736f
C34 clk out_8 0.15291f
C35 clk_bar Din 0.08985f
C36 clk_bar out_4 0.12132f
C37 out_6 out_4 0
C38 out_5 out_2 0.00103f
C39 out_5 out_3 0.1143f
C40 out_7 out_3 0
C41 out_1 vdd 0.42886f
C42 clk_bar clk 1.48592f
C43 clk_bar out_8 0.10669f
C44 out_6 clk 0.13656f
C45 out_6 out_8 0.52027f
C46 Din out_2 0.00304f
C47 Din out_3 0
C48 out_4 out_2 0.51799f
C49 out_4 out_3 0.12613f
C50 clk out_2 0.2121f
C51 clk out_3 0.04785f
C52 out_6 clk_bar 0.18465f
C53 out_7 gnd 1.2882f
C54 Din gnd 0.60951f
C55 clk gnd 3.5342f
C56 vdd gnd 6.02764f
C57 out_8 gnd 0.28851f
C58 out_6 gnd 1.04338f
C59 out_5 gnd 0.56774f
C60 out_4 gnd 0.27511f
C61 out_3 gnd 1.71677f
C62 out_1 gnd 0.44766f
C63 clk_bar gnd 4.09945f
C64 out_2 gnd 0.7627f
.ends

"dff_neg_edge_pex.spice" 160L, 7631B
```

DFF PEX

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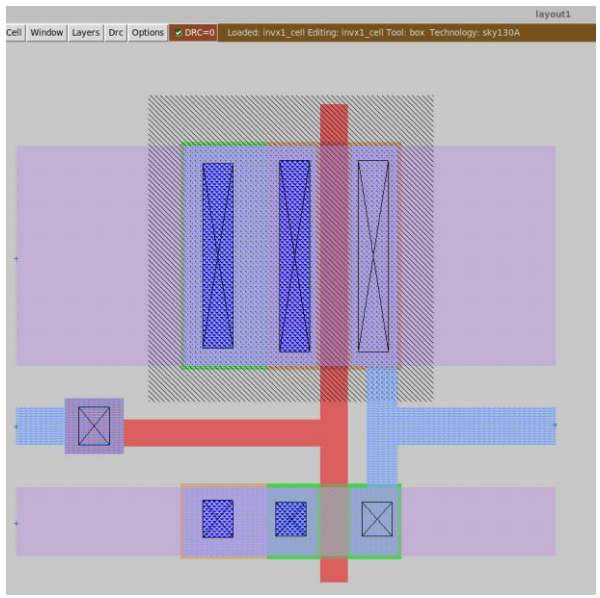
4. DRC and LVS Clean Screenshots

DRC and LVS Clean Screenshots.

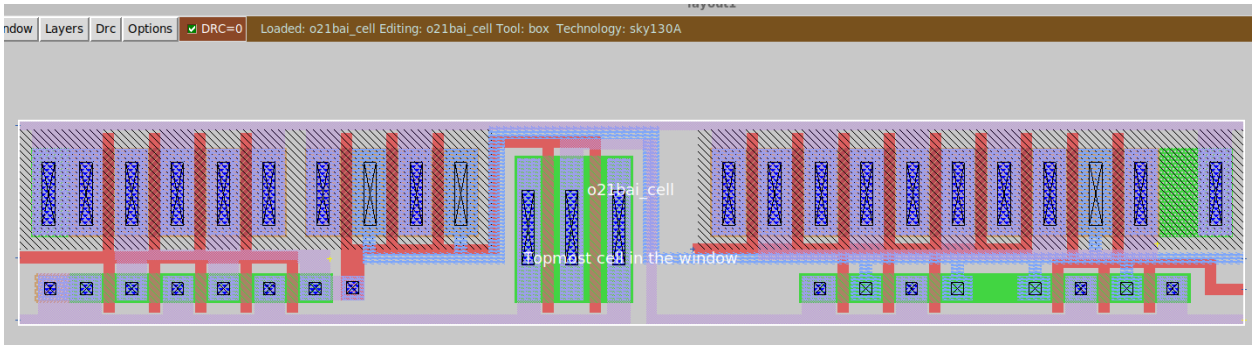
4.1 Combinational Circuit



INVX1 DRC Clean

```
Contents of circuit 1:  Circuit: 'invx1_cell'
Circuit invx1_cell contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   1
  Class: sky130_fd_pr__pfet_01v8 instances:   1
Circuit contains 4 nets.
Contents of circuit 2:  Circuit: 'invx1'
Circuit invx1 contains 2 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   1
  Class: sky130_fd_pr__pfet_01v8 instances:   1
Circuit contains 4 nets.

Circuit 1 contains 2 devices, Circuit 2 contains 2 devices.
Circuit 1 contains 4 nets,    Circuit 2 contains 4 nets.


Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
```

INVX1 LVS

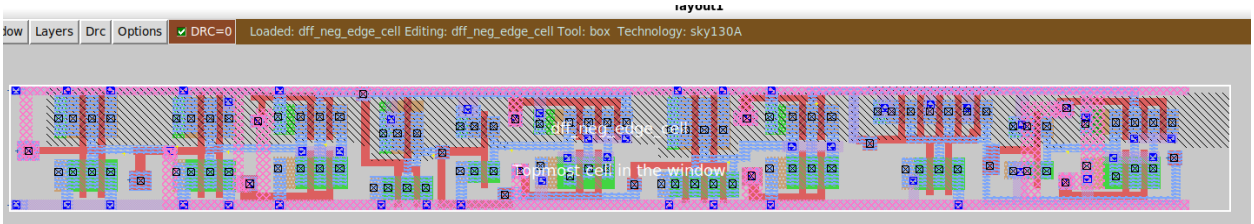


o21bai DRC Clean

```
Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'o21bai_cell'
Circuit o21bai_cell contains 8 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   4
  Class: sky130_fd_pr__pfet_01v8 instances:   4
Circuit contains 9 nets.
Contents of circuit 2:  Circuit: 'o21bai'
Circuit o21bai contains 8 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   4
  Class: sky130_fd_pr__pfet_01v8 instances:   4
Circuit contains 9 nets.

Circuit 1 contains 8 devices, Circuit 2 contains 8 devices.
Circuit 1 contains 9 nets,    Circuit 2 contains 9 nets.


Final result:
Circuits match uniquely.
Property errors were found.

The following cells had property errors:
 o21bai_cell

Logging to file "comp.out" disabled
LVS Done.
```

o21bai LVS

4.2 Sequential Circuit



DFF DRC Clean

```
Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'dff_neg_edge_cell'
Circuit dff_neg_edge_cell contains 22 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  11
  Class: sky130_fd_pr__pfet_01v8 instances:  11
Circuit contains 13 nets.
Contents of circuit 2:  Circuit: 'dff'
Circuit dff contains 22 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  11
  Class: sky130_fd_pr__pfet_01v8 instances:  11
Circuit contains 13 nets.

Circuit 1 contains 22 devices, Circuit 2 contains 22 devices.
Circuit 1 contains 13 nets,    Circuit 2 contains 13 nets.


Final result:
Circuits match uniquely.
Property errors were found.

The following cells had property errors:
 dff_neg_edge_cell

Logging to file "comp.out" disabled
LVS Done.
```

DFF LVS

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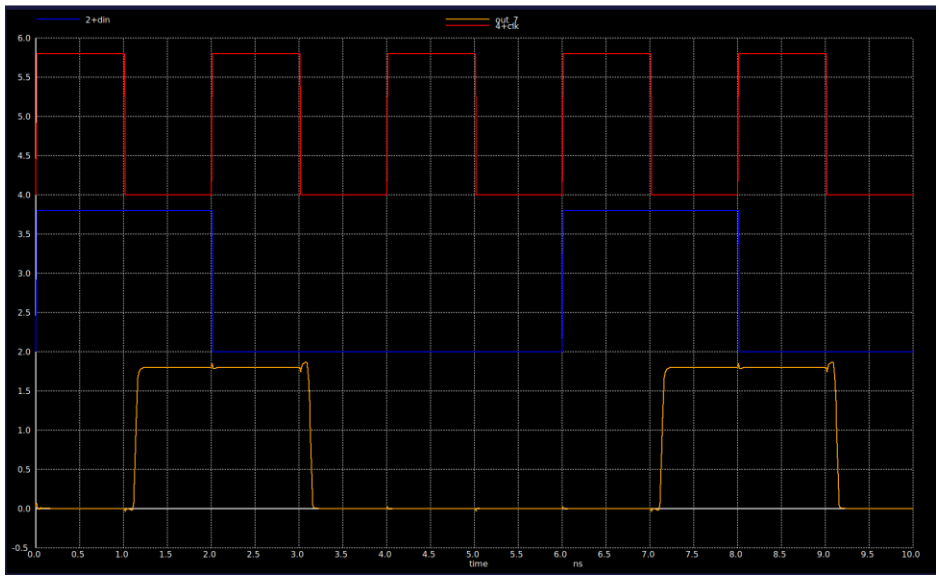
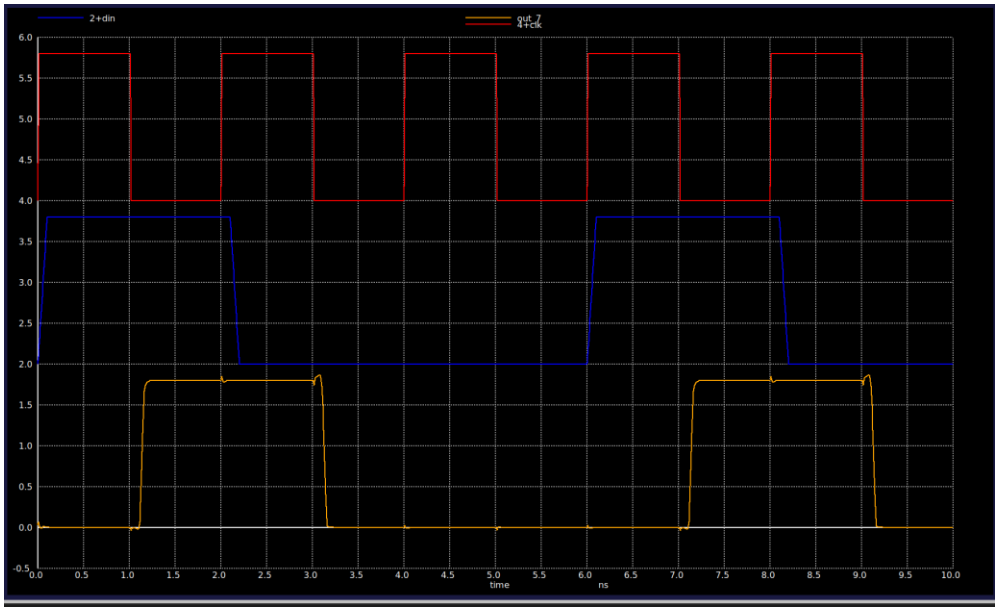
5. Waveforms from one Simulation

5.1 Propagation Delay



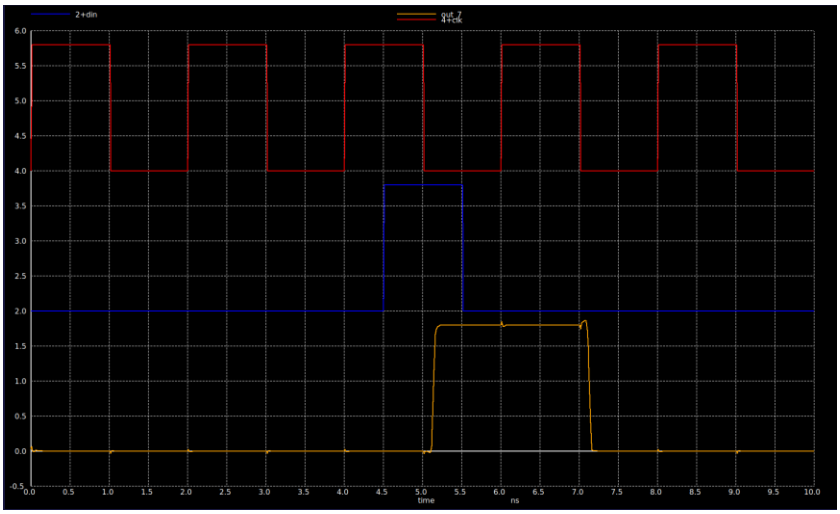
DFF Propagation Delay

5.2 Rise and Fall Transition Time



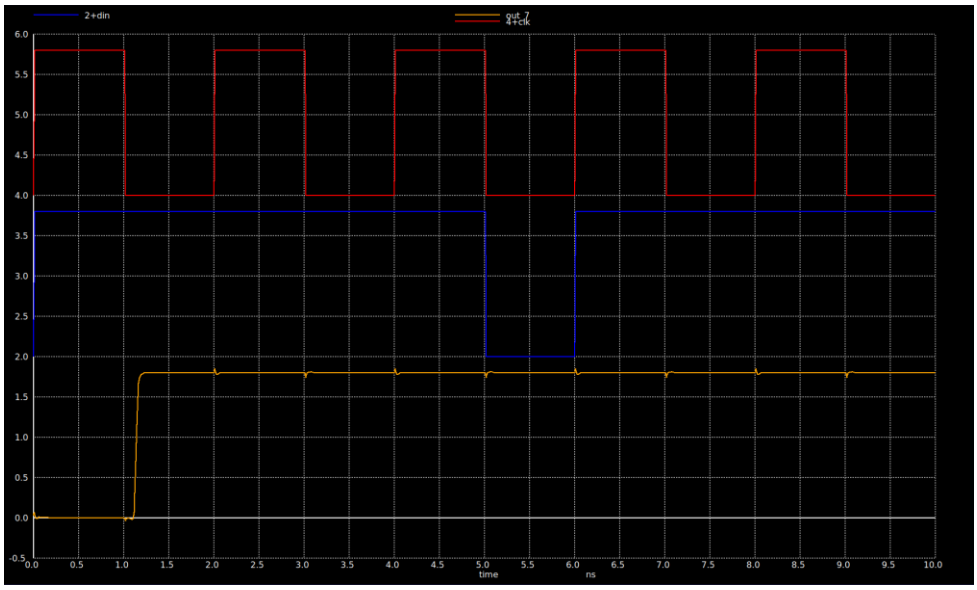
Rise and Fall Transition Time

5.3 Setup Time



Setup Time

5.4 Hold Time



Hold Time

---

6. Timing and Power Tables

A filled Timing and Power tables (similar to the .doc files in the LIB folder).

Please find the attached .docx for more details.

For Inverter (Invx1)

a) **Transition Time Table:** (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ps) [Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	13.7631	24.2355	12.4108
10 fF	67.9552	68.2797	76.3479
100 fF	592.795	592.787	648.462

(ii) Output Fall Transitions (in ps) [Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	14.4664	22.0466	82.3754
10 fF	73.1739	73.0868	175.157
100 fF	641.808	641.829	NA

b) **Propagation delay time tables**:

(i) Cell Rise Delay (in ps) [Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	14.2344	31.9308	12.7670
10 fF	52.3964	74.8147	53.0894
100 fF	336.027	374.913	NA

(ii) Cell Fall Delay (in ps) [Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	18.0980	33.8662	75.1773
10 fF	65.0585	84.4503	214.787
100 fF	500.755	521.081	726.877

c) **Static Power (cover all possible input combinations based on number of inputs).**

Condition (b1_n)	Power (nW)
0	0.0033
1	0.113

d) **Dynamic Power Table:**

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	857.56	750.9	1135.3
10 fF	3992	3921.9	4026.3
100 fF	33163.69	33154.2	33126.9

(ii) Fall Power (in nW) [Input slew vs output capacitance].

[Input slew vs output capacitance].

Related pin b1_n: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	861.5	750	1135.3
10 fF	3996.1	3922.3	4026.3
100 fF	33167.1	33154.4	33127.4

For o21bai Logic

For combinations cells (depending on number of inputs – example, inverter has only A input), the following characterizations have to be performed and filled. Remove all unwanted rows.

e) **Transition Time Table:** (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ps) [Input slew vs output capacitance].

Related pin a1: (i.e., other input pins are held constant)

a2=0

b1_n=0

	10 ps	100 ps	1000 ps
0.5 fF	34.5	36.8	34.40
10 fF	68.33	68.8	68.8
100 fF	352.8	392.8	-

Related pin a2: (i.e., other input pins are held constant)

a1=0

b1_n=0

	10 ps	100 ps	1000 ps
0.5 fF	34.3	39.1	35.8
10 fF	68.3	68.6	76
100 fF	392	392	-

Related pin b1_n: (i.e., other input pins are held constant)

a1=1

a2=1

	10 ps	100 ps	1000 ps
0.5 fF	28.2	28.2	37.3
10 fF	45.2	45.2	51.9
100 fF	215	215	216

(ii) Output Fall Transitions (in ps) [Input slew vs output capacitance].

Related pin a1: (i.e., other input pins are held constant)

a2=0

b_in=0

	10 ps	100 ps	1000 ps
0.5 fF	34.3	34.0	107
10 fF	55.7	54.7	141.7
100 fF	257.9	258	314.7

Related pin a2: (i.e., other input pins are held constant)

a1=0

b_in=0

	10 ps	100 ps	1000 ps
0.5 fF	21.4	26.7	102
10 fF	42.5	44.4	139.1
100 fF	245.1	245	311.4

Related pin b1_n: (i.e., other input pins are held constant)

a1=1

a2=1

	10 ps	100 ps	1000 ps
0.5 fF	15.1	15.3	15.6
10 fF	26	25.9	25.9
100 fF	140	140	140

f) Propagation delay time tables: (unlike textbook definitions that we used for our assignments, here we will use 50% of input to 50% of output to simulate propagation delay – by keeping other inputs fixed).

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin a1: (i.e., other input pins are held constant)

a2=0

b1_n=0

	10 ps	100 ps	1000 ps
0.5 fF	39.4	57.1	37.2
10 fF	59.3	81.4	60.8
100 fF	287.6	306.8	305.1

Related pin a2: (i.e., other input pins are held constant)

a1=0

b1_n=0

	10 ps	100 ps	1000 ps
0.5 fF	30.4	50.3	29.6
10 fF	54.4	74.2	55.4
100 fF	279.1	300	-

Related pin b1_n: (i.e., other input pins are held constant)

a1=1
a2=1

	10 ps	100 ps	1000 ps
0.5 fF	1049	977	645
10 fF	1062	990	660
100 fF	1181	1109	779

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].
Related pin a1 : (i.e., other input pins are held constant)
a2=0
b_in=0

	10 ps	100 ps	1000 ps
0.5 fF	39.8	56.4	151
10 fF	63.1	75.5	195
100 fF	220.1	236.8	437.2

Related pin a2: (i.e., other input pins are held constant)
a1=0
b_in=0

	10 ps	100 ps	1000 ps
0.5 fF	23.6	41.2	91.1
10 fF	40.2	59.9	147.8
100 fF	196.2	216.5	414

Related pin b1_n: (i.e., other input pins are held constant)
a1=1
a2=1

	10 ps	100 ps	1000 ps
0.5 fF	1058	1169	1547
10 fF	1068	1180	1557
100	1157	1269	1646

g) Static Power (cover all possible input combinations based on number of inputs).

Condition (b1_n a2 a1)	Power (nW)
000	0.011
001	1.46
010	1.64
011	0.85
100	0.79
101	0.808
110	0.808
111	0.808

h) Dynamic Power Table:
(i) Rise Power (in nW) [Input slew vs output capacitance].
Related pin a1: (i.e., other input pins are held constant)
a2=0
b_in=0

	10 ps	100 ps	1000 ps
0.5 fF	5776.7	5657.4	5877.8
10 fF	8877.4	8774.7	8901.0
100 fF	38058.4	38027.4	36995.8

Related pin a2: (i.e., other input pins are held constant)
a1=0
b_n=0

	10 ps	100 ps	1000 ps
0.5 fF	2652.7	2534.1	3077.2
10 fF	5761.3	5652.4	6049.5
100 fF	34946	34916.1	33270.7

Related pin b1_n: (i.e., other input pins are held constant)
a1=1
a2=1

	10 ps	100 ps	1000 ps
0.5 fF	5136.6	4795.2	5276.02
10 fF	8273.9	7923.6	8355.1
100 fF	37530.7	37211.6	37590.4

(ii) Fall Power (in nW) [Input slew vs output capacitance].
Related pin a1: (i.e., other input pins are held constant)
a2=0
b_n=0

	10 ps	100 ps	1000 ps
0.5 fF	735.8	263.5	359.8
10 fF	739.7	306.7	296.5
100 fF	740.1	383.7	197.9

Related pin a2: (i.e., other input pins are held constant)
a1=0
b_n=0

	10 ps	100 ps	1000 ps
0.5 fF	884.6	751.3	976.2
10 fF	904.3	814.6	871.9
100 fF	920.3	914.22	839.3

Related pin b1_n: (i.e., other input pins are held constant)
a1=1
a2=1

	10 ps	100 ps	1000 ps
0.5 fF	7445.6	7195.3	8215.7
10 fF	7523.2	7272.5	8216.1
100 fF	7602.2	7396.6	8429.5

For DFF Sequential Circuit

a) **Set-up Time Table:**

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.0435	0.15
1000 ps	0.12	0.207

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.08	0.27
1000 ps	0.51	0.22

b) Hold Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.07	0.25
1000 ps	0.02	0.14

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.085	0.23
1000 ps	0.19	0.20

c) Transition Time Table (for Q output, CLK will have minimum slew of 10 ps): (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0269	0.02696	0.02677
10 fF	0.04876	0.04876	0.049068
1–00 fF	0.028555	0.028555	0.028555

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.02934	0.02931	0.02930
10 fF	0.046342	0.046923	0.046591
100 fF	0.018166	0.018171	0.01817

d) CLK-to-Q Delay Time Table: (delay between clock transition and data transition. Use 50% of CLK to 50% of output to simulate propagation delay).

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.121313	0.121313	0.12099
10 fF	0.14154	0.14153	0.14126
100 fF	0.30745	0.30743	0.30719

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.11481	0.114757	0.114747

10 fF	0.13791	0.137848	0.137847
100 fF	0.267972	0.267844	0.26784

e) Static Power (<u>all possible input combinations</u> of CLK and D).

Condition (CLK, D)	Power (nW)
00	156859.1
01	156859.2
10	1.61
11	2.1

f) Dynamic Power Table: (CLK will have minimum slew of 10 ps)

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	18028.7	19417.6	23623.9
10 fF	21138.4	22527.1	26733.4
100 fF	50420.7	51808.2	56015.2

(ii) Fall Power (in nW) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	18623.4	18240.6	18646.6
10 fF	18713.5	18330.9	18736.7
100 fF	18881.7	18499.1	18904.9

7. Table of Contributions

Contribution Summary Information

Name	Roll No.	
Amit Arvind Kale	25D0545	Capacitance Calculation (Comb + Seq) + Lib file Calculations
Antriksh	25M1205	Sequential Schematic + Layout + Lib file Calculations
Karunesh Kumar Tripathi	25D0529	Combinational Schematic + Combinational Layout + Lib file calculations +
Tejas Sunita Anil Hande	25D0537	Combinational Schematic + + Verilog Programs + Report Writeup